

Begum Rokeya University, Rangpur



Assignment On Categorical Data Analysis(CDA)

Course Title: Categorical Data Analysis

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Submission Date:

Logit Regression

Question 1. A researcher is interested in how variables, such as GRE (Graduate Record Exam scores), GPA (grade point average) and prestige of the undergraduate institution, effect admission into graduate school. The response variable, admit/don't admit, is a binary variable.

R code and results:

```
setwd("D:/Courses of Siddik sir/4104")
getwd()

## [1] "D:/Courses of Siddik sir/4104"

install.packages("aod")
install.packages("ggplot2")

library(aod)
library(ggplot2)
#read data
mydata= read.csv("cda11.csv")

## view the first few rows of the data

head(mydata)

##   admit gre  gpa rank
## 1     0 380 3.61    3
## 2     1 660 3.67    3
## 3     1 800 4.00    1
## 4     1 640 3.19    4
## 5     0 520 2.93    4
## 6     1 760 3.00    2
```

Comment: This dataset has four variables, where admit is the response variable and gre, gpa and rank are predictor variables. Here rank is the binary response variable.

R code and results:

```
summary(mydata)
```

	admit	gre	gpa	rank
## Min.	:0.0000	Min. :220.0	Min. :2.260	Min. :1.000
## 1st Qu.	:0.0000	1st Qu.:520.0	1st Qu.:3.130	1st Qu.:2.000
## Median	:0.0000	Median :580.0	Median :3.395	Median :2.000
## Mean	:0.3175	Mean :587.7	Mean :3.390	Mean :2.485

```
## 3rd Qu.:1.0000 3rd Qu.:660.0 3rd Qu.:3.670 3rd Qu.:3.000
## Max. :1.0000 Max. :800.0 Max. :4.000 Max. :4.000
```

Comment: Summary gives us the basic descriptives(Min., 1st Qu., Median, Mean, 3rd Qu., Max.) for each variable, admit, gre, gpa and rank respectfully.

R code and results:

```
# Standard deviation
std = sapply(mydata, sd); std

##      admit      gre      gpa      rank
## 0.4660867 115.5165364 0.3805668 0.9444602
```

Comment: The standard deviation of admit, gre, gpa and rank are 0.466, 115.517, 0.381, 0.944 respectfully.

R code and results:

```
## two-way contingency table of categorical outcome and predictors we want
## to make sure there are not 0 cells

xtabs(~admit+rank, data = mydata)

##      rank
## admit 1  2  3  4
##      0 28 97 93 55
##      1 33 54 28 12

mydata$rank=factor(mydata$rank)
```

Comment: Here we have created a two-way contingency table in terms of admit and rank.

R code and results:

```
mylogit= glm(admit ~ gre + gpa + rank, data = mydata, family = "binomial"
)
```

```
summary(mylogit)
```

```
##
## Call:
## glm(formula = admit ~ gre + gpa + rank, family = "binomial",
##      data = mydata)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -3.989979   1.139951  -3.500 0.000465 ***
## gre          0.002264   0.001094   2.070 0.038465 *
## gpa          0.804038   0.331819   2.423 0.015388 *
## rank2       -0.675443   0.316490  -2.134 0.032829 *
## rank3       -1.340204   0.345306  -3.881 0.000104 ***
## rank4       -1.551464   0.417832  -3.713 0.000205 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 499.98  on 399  degrees of freedom
## Residual deviance: 458.52  on 394  degrees of freedom
## AIC: 470.52
##
## Number of Fisher Scoring iterations: 4
```

Comment: The summary result illustrates that, for every one unit change in gre, the log odds of admission (Versus non-admission) increase by 0.002. For a one unit increase in gpa, the log odds of being admitted to graduate school increases by 0.804. The indicator variables for rank illustrates that having attended an undergraduate institution with rank of 2, versus an institution with a rank of 1, changes the log odds of admission by -0.675. Having attended an undergraduate institution with rank of 3, versus an institution with a rank of 1, changes the log odds of admission by -1.340. Similarly, changes of log odds of admission with rank of 4 versus rank of 1 is -1.551.

R code and results:

```
## CIs using profiled log-likelihood
```

```
confint(mylogit)
```

```
## Waiting for profiling to be done...
```

```
##              2.5 %          97.5 %
## (Intercept) -6.2716202334 -1.792547080
## gre          0.0001375921  0.004435874
## gpa          0.1602959439  1.464142727
```

```
## rank2      -1.3008888002 -0.056745722
## rank3      -2.0276713127 -0.670372346
## rank4      -2.4000265384 -0.753542605
```

```
confint.default(mylogit)
```

```
##              2.5 %      97.5 %
## (Intercept) -6.2242418514 -1.755716295
## gre          0.0001202298  0.004408622
## gpa          0.1536836760  1.454391423
## rank2        -1.2957512650 -0.055134591
## rank3        -2.0169920597 -0.663415773
## rank4        -2.3703986294 -0.732528724
```

Comment: The above result presents the confident intervals for each variables. 95% confidence intervals for intercept = [-6.27, -1.79], 95% confidence intervals for gre= [0.00014, 0.0044], 95% confidence intervals for gpa=[0.160, 1.464], 95% confidence intervals for rank2 in terms of rank1=[-1.296, -0.055], 95% confidence intervals for rank3 in terms of rank1= [-2.017, -0.663], 95% confidence intervals for rank4 in terms of rank1=[-2.370, -0.732].

R code and results:

```
wald.test(b= coef(mylogit), Sigma = vcov(mylogit), Terms = 4:6)
```

```
## Wald test:
## -----
##
## Chi-squared test:
## X2 = 20.9, df = 3, P(> X2) = 0.00011
```

Comment: Here the wald test is done to test the overall effect of rank. The chi-squared test statistic of 20.9, with three degrees of freedom is associated with a p-value of 0.00011 indicating that the overall effect of rank is statistically significant.

R code and results:

```
v = cbind(0,0,0,1,-1,0)
wald.test(b= coef(mylogit), Sigma = vcov(mylogit), L = v)

## Wald test:
## -----
##
## Chi-squared test:
## X2 = 5.5, df = 1, P(> X2) = 0.019
```

Comment: This above test is done to test the hypotheses about the differences in the coefficients for the different levels of rank. The chi-squared test statistic of 5.5 with 1 degree of freedom is associated with a p-value of 0.019, indicating that the difference between the coefficient for rank=2 and the coefficient for rank=3 is statistically significant

R code and results:

```
# odds ratios only

exp(coef(mylogit))

## (Intercept)      gre      gpa      rank2      rank3      rank4
##  0.0185001    1.0022670  2.2345448  0.5089310  0.2617923  0.2119375

# odds ratios and 95% C
exp(cbind(OR= coef(mylogit), confint(mylogit)))

## Waiting for profiling to be done...

##              OR      2.5 %    97.5 %
## (Intercept) 0.0185001 0.001889165 0.1665354
## gre         1.0022670 1.000137602 1.0044457
## gpa         2.2345448 1.173858216 4.3238349
## rank2       0.5089310 0.272289674 0.9448343
## rank3       0.2617923 0.131641717 0.5115181
## rank4       0.2119375 0.090715546 0.4706961
```

Comment: We have exponentiated the coefficients and confident intervals to get the odds ratios. From the above results we can say we can say that for a one unit increase in gpa, the odds of being admitted to graduate school (versus not being admitted) increase by a factor of 2.23.

R code and results:

```
#create new data frame
newdata1= with(mydata, data.frame(gre= mean(gre), gpa= mean(gpa), rank= fac
tor(1:4)));newdata1

##      gre      gpa rank
## 1 587.7 3.3899    1
## 2 587.7 3.3899    2
## 3 587.7 3.3899    3
## 4 587.7 3.3899    4

newdata1$rankP= predict(mylogit, newdata= newdata1, type= "response");newd
ata1$rankP
```

```
## [1] 0.5166016 0.3522846 0.2186120 0.1846684

newdata1

##      gre      gpa rank      rankP
## 1 587.7 3.3899     1 0.5166016
## 2 587.7 3.3899     2 0.3522846
## 3 587.7 3.3899     3 0.2186120
## 4 587.7 3.3899     4 0.1846684
```

Comment: The above output represents predicted probability of being accepted into a graduate program is 0.52 for students from the highest prestige undergraduate institutions(rank=1) . Similarly for rank2, rank3, rank4, the predicted probabilities are 0.35, 0.21 and 0.18.

R code and results:

```
newdata2= with(mydata, data.frame(gre= rep(seq(from= 200, to=800,length.out= 100), 4),gpa= mean(gpa),rank=factor(rep(1:4,each= 100))))

newdata3 = cbind(newdata2, predict(mylogit,newdata= newdata2, type= "link", se= TRUE))

newdata3= within(newdata3,{
  predictedProb= plogis(fit)
  LL= plogis(fit - (1.96* se.fit))
  UL= plogis(fit + (1.96* se.fit))
})
head(newdata3)
```

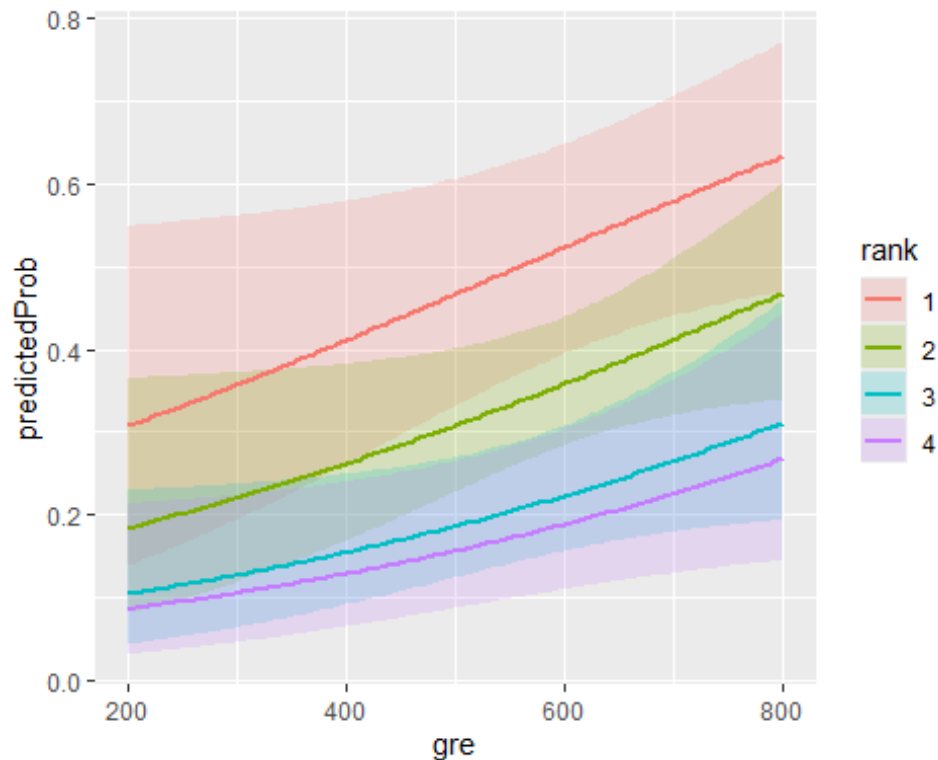
##	gre	gpa	rank	fit	se.fit	residual.scale	UL
LL predictedProb							
## 1	200.0000	3.3899	1	-0.8114870	0.5147714	1	0.5492064 0.
	1393812	0.3075737					
## 2	206.0606	3.3899	1	-0.7977632	0.5090986	1	0.5498513 0.
	1423880	0.3105042					
## 3	212.1212	3.3899	1	-0.7840394	0.5034491	1	0.5505074 0.
	1454429	0.3134499					
## 4	218.1818	3.3899	1	-0.7703156	0.4978239	1	0.5511750 0.
	1485460	0.3164108					
## 5	224.2424	3.3899	1	-0.7565919	0.4922237	1	0.5518545 0.
	1516973	0.3193867					
## 6	230.3030	3.3899	1	-0.7428681	0.4866494	1	0.5525464 0.
	1548966	0.3223773					

Comment: Here we've created predicted probabilities varying the value of gre and rank. In convenience of plotting we've created create 100 values of gre between 200 and 800, at each

value of rank (i.e., 1, 2, 3, and 4). So the predicted probabilities are 0.32, 0.31, 0.313, 0.32 etc .

R code and results: Graphical representation

```
ggplot(newdata3, aes(x= gre, y= predictedProb)) + geom_ribbon(aes(ymin = L,
    ymax= UL, fill = rank), alpha = 0.2) + geom_line(aes(colour= rank)
, size= 1)
```



Comment: Here we've plotted the gre score and the predicted probabilities. Where ,100 predicted probabilities for 100 gre scores for 4 different ranks(rank1, rank2, rank3 and rank4) are respresented graphically. The plot illustrates that the higher the gre number is, the higher the predicted probability of getting admission in different ranked universities.

R code and results:

```
with(mylogit,null.deviance - deviance)
## [1] 41.45903
with(mylogit, null.deviance - deviance)
## [1] 41.45903
```

```

with(mylogit, df.null - df.residual)

## [1] 5

with(mylogit, pchisq(null.deviance - deviance, df.null - df.residual, lower.tail = FALSE))

## [1] 7.578194e-08

logLik(mylogit)

## 'log Lik.' -229.2587 (df=6)

```

Comment: The chi-square of 41.46 with 5 degrees of freedom and an associated p-value of less than 0.001 tells us that our model as a whole fits significantly better than an empty model.

Poisson Regression

Question 2: The number of awards earned by students at one high school. Predictors of the number of awards earned include the type of program in which the student was enrolled (e.g., vocational, general or academic) and the score on their final exam in math.

R code and results:

```

##poisson Regression
install.packages("sandwich")

install.packages("msm")

library(ggplot2)
library(sandwich)
library(msm)
#read data
p= read.csv("https://stats.idre.ucla.edu/stat/data/poisson_sim.csv")

```

```
p= within(p, {
  prog= factor(prog,levels=1:3, labels = c("General", "Academic","Vocation
al"))
  id= factor(id)
})
summary(p)
```

##	id	num_awards	prog	math
## 1	: 1	Min. :0.00	General : 45	Min. :33.00
## 2	: 1	1st Qu.:0.00	Academic :105	1st Qu.:45.00
## 3	: 1	Median :0.00	Vocational: 50	Median :52.00
## 4	: 1	Mean :0.63		Mean :52.65
## 5	: 1	3rd Qu.:1.00		3rd Qu.:59.00
## 6	: 1	Max. :6.00		Max. :75.00
##	(Other):194			

Comment: The above results represents some descriptive statistics of 200 students in a school. There are 45 students in general program, 105 students in academic program and 50 students in vocational program. Among the students, the minimum number of award is 0 and maximum is 6. The minimum score on their final exam on math is 33.00 and maximum score is 75.

R code and results:

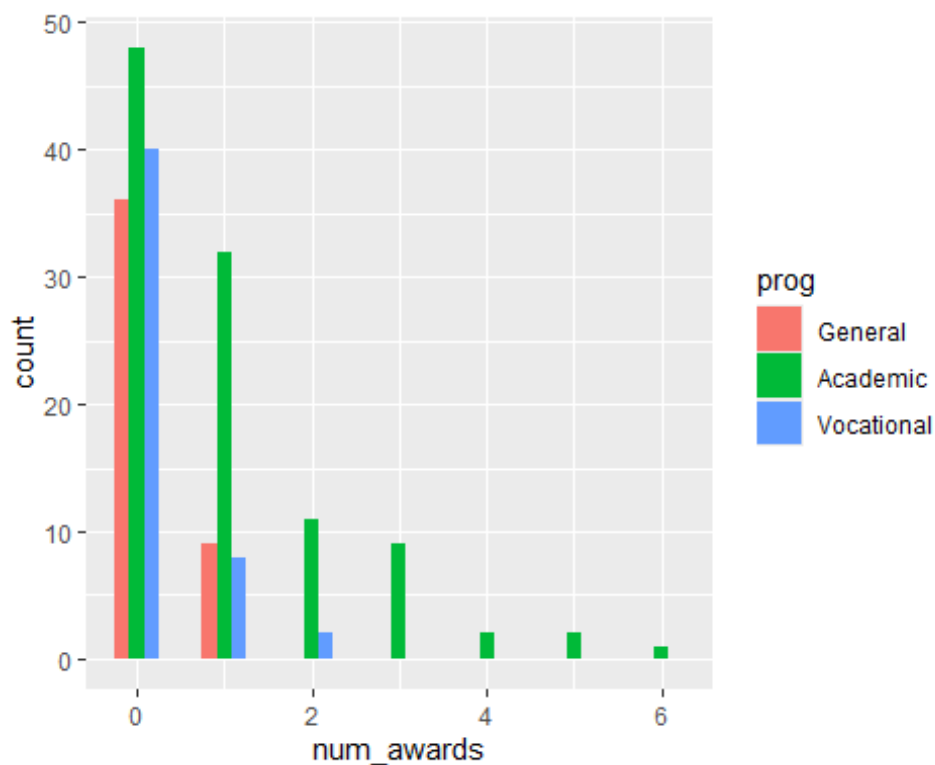
```
with(p,tapply(num_awards, prog, function(x){
  sprintf("M (SD) = %1.2f (%1.2f)", mean(x), sd(x))
}))
```

##	General	Academic	Vocational
##	"M (SD) = 0.20 (0.40)"	"M (SD) = 1.00 (1.28)"	"M (SD) = 0.24 (0.52)"

Comment: The above result shows the average numbers of awards by program type and seems to suggest that program type is a good candidate for predicting the number of awards, our outcome variable, because the mean value of the outcome appears to vary by program. Hence we can say, students of academic program get more number of awards than the students of general and vocational program .

R code and results: Graphical representation

```
ggplot(p, aes(num_awards, fill= prog))+
  geom_histogram(binwidth = .5, position = "dodge")
```



Comment: The above plot represents the number of awards according to different programs with their count respectively. Basically, a conditional histogram separated out by program type is plotted to show the distribution. The overall graph presents that, the academic program students get more awards than other program students. Then in second position there is vocational program students and then general program in third position.

R code and results:

```
m1 = glm(num_awards ~ prog+ math, family = "poisson",data = p);m1
##
## Call:  glm(formula = num_awards ~ prog + math, family = "poisson", data
= p)
##
## Coefficients:
## (Intercept)    progAcademic  progVocational         math
##    -5.24712         1.08386         0.36981         0.07015
##
## Degrees of Freedom: 199 Total (i.e. Null);  196 Residual
```

```
## Null Deviance:      287.7
## Residual Deviance: 189.4      AIC: 373.5

summary(m1)

##
## Call:
## glm(formula = num_awards ~ prog + math, family = "poisson", data = p)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -5.24712    0.65845  -7.969 1.60e-15 ***
## progAcademic    1.08386    0.35825   3.025  0.00248 **
## progVocational  0.36981    0.44107   0.838  0.40179
## math           0.07015    0.01060   6.619 3.63e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 287.67  on 199  degrees of freedom
## Residual deviance: 189.45  on 196  degrees of freedom
## AIC: 373.5
##
## Number of Fisher Scoring iterations: 6
```

Comment: The above outcome illustrates, the students in the progAcademic program, on average, are expected to receive about 1.1 times as many awards as students in the general group. This means students in the academic program tend to receive more awards compared to those in the general program, after accounting for other factors like math scores. Students in the progVocational program, on average, are expected to receive about 0.37 times as many awards as students in the reference group (prog = General). This indicates that students in the vocational program tend to receive fewer awards compared to those in the general program, after considering other factors like math scores. For every one-unit increase in the math score, the expected number of num_awards increases by approximately 0.07. So, if a student's math score goes up by 1 point, we expect their number of awards to increase by about 0.07 on average, assuming all else remains the same.

R code and results:

```
cov.m1 <- vcovHC(m1, type="HC0")
std.err <- sqrt(diag(cov.m1))
r.est <- cbind(Estimate= coef(m1), "Robust SE" = std.err,
               "Pr(>|z|)" = 2 * pnorm(abs(coef(m1)/std.err), lower.tail=FALSE),
               LL = coef(m1) - 1.96 * std.err,
               UL = coef(m1) + 1.96 * std.err)
```

```

r.est
##              Estimate Robust SE      Pr(>|z|)          LL
UL
## (Intercept)   -5.2471244 0.64599839 4.566630e-16 -6.51328124 -3.980967
56
## progAcademic    1.0838591 0.32104816 7.354745e-04  0.45460476  1.713113
53
## progVocational  0.3698092 0.40041731 3.557157e-01 -0.41500870  1.154627
16
## math           0.0701524 0.01043516 1.783975e-11  0.04969947  0.090605
32

with(m1, cbind(res.deviance = deviance, df = df.residual,
               p = pchisq(deviance, df.residual, lower.tail=FALSE)))

##      res.deviance  df      p
## [1,]      189.4496 196 0.6182274

```

Comment: The residual deviance to perform a goodness of fit test for the overall model. Therefore, if the residual difference is small enough, the goodness of fit test will not be significant, indicating that the model fits the data. We conclude that the model fits reasonably well because the goodness-of-fit chi-squared test is not statistically significant because the p value is greater than 0.05.

R code and results:

```

## update m1 model dropping prog
m2 <- update(m1, . ~ . - prog)

## test model differences with chi square test
anova(m2, m1, test="Chisq")

## Analysis of Deviance Table
##
## Model 1: num_awards ~ math
## Model 2: num_awards ~ prog + math
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1      198      204.02
## 2      196      189.45  2    14.572 0.0006852 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Comment: The above analysis of variance table represents the overall effect of prog by comparing the deviance of the full model with the deviance of the model excluding prog. The two degree-of-freedom chi-square test indicates that prog+math is a statistically significant predictor of num_awards.

R code and results:

```
s <- deltamethod(list(~ exp(x1), ~ exp(x2), ~ exp(x3), ~ exp(x4)),
                  coef(m1), cov.m1)
## exponentiate old estimates dropping the p values
rexp.est <- exp(r.est[, -3])
## replace SEs with estimates for exponentiated coefficients
rexp.est[, "Robust SE"] <- s

rexp.est

##              Estimate Robust SE          LL          UL
## (Intercept)  0.00526263 0.00339965 0.001483604 0.01866757
## progAcademic  2.95606545 0.94903937 1.575550534 5.54620292
## progVocational 1.44745846 0.57958742 0.660334536 3.17284023
## math          1.07267164 0.01119351 1.050955210 1.09483681
```

Comment: Students in the academic program are almost three times more likely to receive awards compared to those in the general program. Students in the vocational program are about 1.45 times more likely to receive awards compared to those in the general program. For every one-point increase in math scores, students are 7% more likely to receive awards.

R code and results:

```
(s1 <- data.frame(math = mean(p$math),
                  prog = factor(1:3, levels = 1:3, labels = levels(p$prog))
)))

##      math      prog
## 1 52.645   General
## 2 52.645   Academic
## 3 52.645 Vocational

predict(m1, s1, type="response", se.fit=TRUE)

## $fit
##      1      2      3
## 0.2114109 0.6249446 0.3060086
##
## $se.fit
##      1      2      3
## 0.07050108 0.08628117 0.08833706
##
## $residual.scale
## [1] 1
```

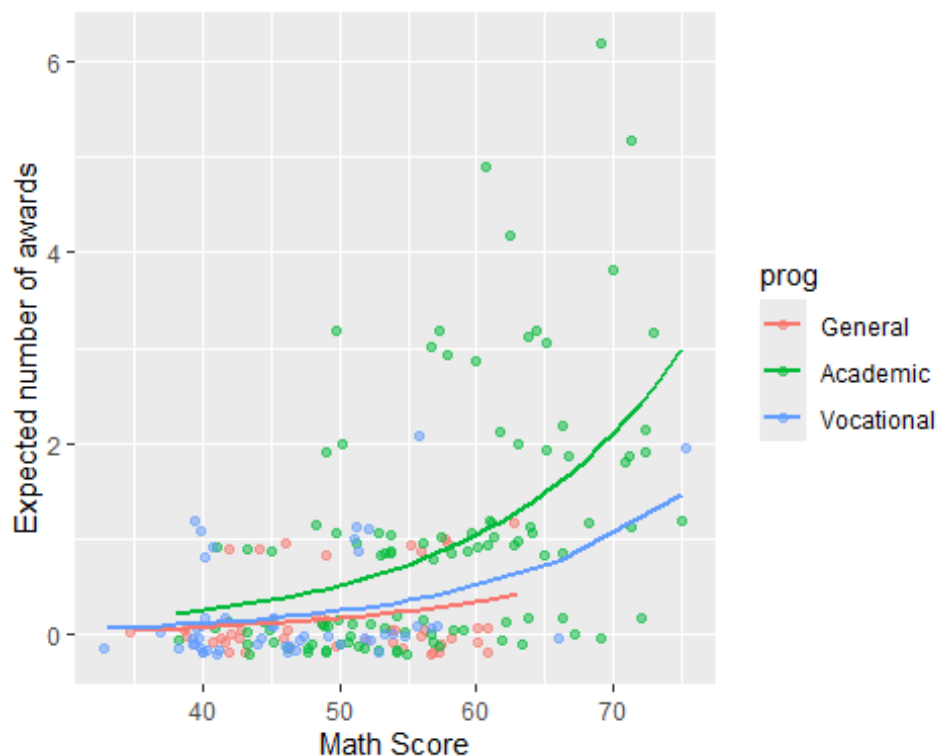
Comment: In the output above, we see that the predicted number of events for level 1 of prog is about .21, holding math at its mean. The predicted number of events for level 2 of prog is higher at .62, and the predicted number of events for level 3 of prog is about .31.

R code and results:

```
## calculate and store predicted values
p$phat <- predict(m1, type="response")

## order by program and then by math
p <- p[with(p, order(prog, math)), ]

## create the plot
ggplot(p, aes(x = math, y = phat, colour = prog)) +
  geom_point(aes(y = num_awards), alpha=.5, position=position_jitter(h=.2)) +
  geom_line(size = 1) +
  labs(x = "Math Score", y = "Expected number of awards")
```



Comment: The above graph indicates that the most awards are predicted for those in the academic program (prog = 2), especially if the student has a high math score. The lowest number of predicted awards is for those students in the general program (prog = 1). The graph overlays the lines of expected values onto the actual points, although a small amount of random noise was added vertically to lessen overplotting.

Negative Binomial Regression

Question 3: School administrators study the attendance behavior of high school juniors at two schools. Predictors of the number of days of absence include the type of program in which the student is enrolled and a standardized test in math.

R code and results:

```
#negative binomial
install.packages("foreign")

install.packages("MASS")

library(ggplot2)
library(foreign)
library(MASS)

dat <- read.dta("https://stats.idre.ucla.edu/stat/stata/dae/nb_data.dta")
dat <- within(dat, {
  prog <- factor(prog, levels = 1:3, labels = c("General", "Academic", "Vocational"))
  id <- factor(id)
})

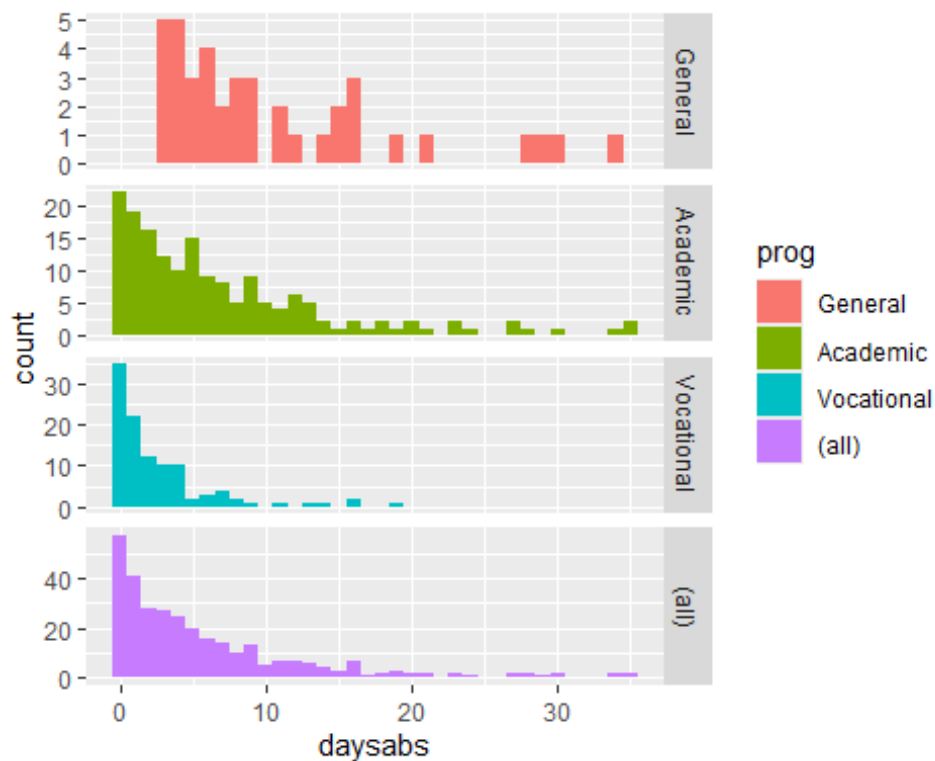
summary(dat)
```

##	id	gender	math	daysabs	pr
og					
## 1001	: 1	female:160	Min. : 1.00	Min. : 0.000	General
: 40					
## 1002	: 1	male :154	1st Qu.:28.00	1st Qu.: 1.000	Academic
:167					
## 1003	: 1		Median :48.00	Median : 4.000	Vocational
:107					
## 1004	: 1		Mean :48.27	Mean : 5.955	
## 1005	: 1		3rd Qu.:70.00	3rd Qu.: 8.000	
## 1006	: 1		Max. :99.00	Max. :35.000	
## (Other):308					

Comment: The above results represents some descriptive statistics of 314 high school juniors from two urban high schools. There are 40 students in general program, 167 students in academic program and 107 students in vocational program. Among the students, 160 are female and 154 are male. The minimum number in math is 1 and maximum is 99. The minimum days of absence is 0 and maximum score is 35.

R code and results: Graphical representation

```
ggplot(dat, aes(daysabs, fill = prog)) + geom_histogram(binwidth = 1) +
  facet_grid(prog ~ ., margins = TRUE, scales = "free")
```



Comment: Here we've plotted the daysabs in x axis and the count of students in y axis. First three plots represents count of students in terms of daysabs separately for the different programs. The last plot count the students for all the programs together.

R code and results:

```
#
with(dat, tapply(daysabs, prog, function(x) {
  sprintf("M (SD) = %1.2f (%1.2f)", mean(x), sd(x))
}))

##                General                Academic                Vocational
## "M (SD) = 10.65 (8.20)" "M (SD) = 6.93 (7.45)" "M (SD) = 2.67 (3.73)"
```

Comment: The table above shows the average numbers of days absent by program type and seems to suggest that program type is a good candidate for predicting the number of days absent, our outcome variable, because the mean value of the outcome appears to vary by program. The variances within each level of prog are higher than the means within each level. These are the conditional means and variances.

These differences suggest that over-dispersion is present and that a **Negative Binomial model** would be appropriate.

R code and results:

```
summary(m1 <- glm.nb(daysabs ~ math + prog, data = dat))

##
## Call:
## glm.nb(formula = daysabs ~ math + prog, data = dat, init.theta = 1.0327
13156,
##      link = log)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    2.615265    0.197460  13.245 < 2e-16 ***
## math          -0.005993    0.002505  -2.392  0.0167 *
## progAcademic  -0.440760    0.182610  -2.414  0.0158 *
## progVocational -1.278651    0.200720  -6.370 1.89e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(1.0327) family taken to be
1)
##
##      Null deviance: 427.54  on 313  degrees of freedom
## Residual deviance: 358.52  on 310  degrees of freedom
## AIC: 1741.3
##
## Number of Fisher Scoring iterations: 1
##
##
##              Theta:  1.033
##             Std. Err.: 0.106
##
## 2 x log-likelihood: -1731.258
```

Comment: The above result illustrates that with each one-point increase in math scores, the expected number of days absent decreases by 0.006. Students in the academic program tend to have 0.44 fewer days absent compared to the general group(reference group). Students in the vocational program tend to have 1.28 fewer days absent compared to the general group(reference group).

R code and results:

```
#ANOVA test
m2 <- update(m1, . ~ . - prog)
anova(m1, m2)

## Likelihood ratio tests of Negative Binomial Models
##
## Response: daysabs
##          Model      theta Resid. df    2 x log-lik.  Test      df LR stat.
Pr(Chi)
## 1          math 0.8558565      312      -1776.306
## 2 math + prog 1.0327132      310      -1731.258 1 vs 2      2 45.04798
1.65179e-10
```

Comment: We have compared a model with and without program to determine if program itself, overall, is statistically significant. The two degree-of-freedom chi-square test indicates that prog is a statistically significant predictor of daysabs.

R code and results: Checking model assumptions

```
m3 <- glm(daysabs ~ math + prog, family = "poisson", data = dat)
pchisq(2 * (logLik(m1) - logLik(m3)), df = 1, lower.tail = FALSE)

## 'log Lik.' 2.157298e-203 (df=5)
```

Comment: the associated chi-squared value estimated from $2 * (\log\text{Lik}(m1) - \log\text{Lik}(m3))$ is 926.03 with one degree of freedom. This strongly suggests the negative binomial model, estimating the dispersion parameter, is more appropriate than the Poisson model.

R code and results:

```
(est <- cbind(Estimate = coef(m1), confint(m1)))

## Waiting for profiling to be done...

##          Estimate      2.5 %      97.5 %
## (Intercept)  2.615265446  2.24205576  3.012935926
## math        -0.005992988 -0.01090086 -0.001066615
## progAcademic -0.440760012 -0.81006586 -0.092643481
## progVocational -1.278650721 -1.68348970 -0.890077623

exp(est)

##          Estimate      2.5 %      97.5 %
## (Intercept)  13.6708448  9.4126616 20.3470498
## math         0.9940249  0.9891583  0.9989340
```

```
## progAcademic      0.6435471 0.4448288 0.9115184
## progVocational    0.2784127 0.1857247 0.4106239
```

Comment: The output above indicates that the incident rate for progAcademic is 0.64 times the incident rate for the reference group (progGeneral). Likewise, the incident rate for ProgVocational is 0.28 times the incident rate for the reference group holding the other variables constant. The percent change in the incident rate of daysabs is a 1% decrease for every unit increase in math.

R code and results: Predicted Values

```
newdata1 <- data.frame(math = mean(dat$math), prog = factor(1:3, levels =
1:3,
                                labels = level
s(dat$prog)))
newdata1$phat <- predict(m1, newdata1, type = "response")
newdata1

##      math      prog      phat
## 1 48.26752   General 10.236899
## 2 48.26752   Academic 6.587927
## 3 48.26752 Vocational 2.850083
```

Comment: In the output above, we see that the predicted number of events (e.g., days absent) for a general program is about 10.24, holding math at its mean. The predicted number of events for an academic program is lower at 6.59, and the predicted number of events for a vocational program is about 2.85.

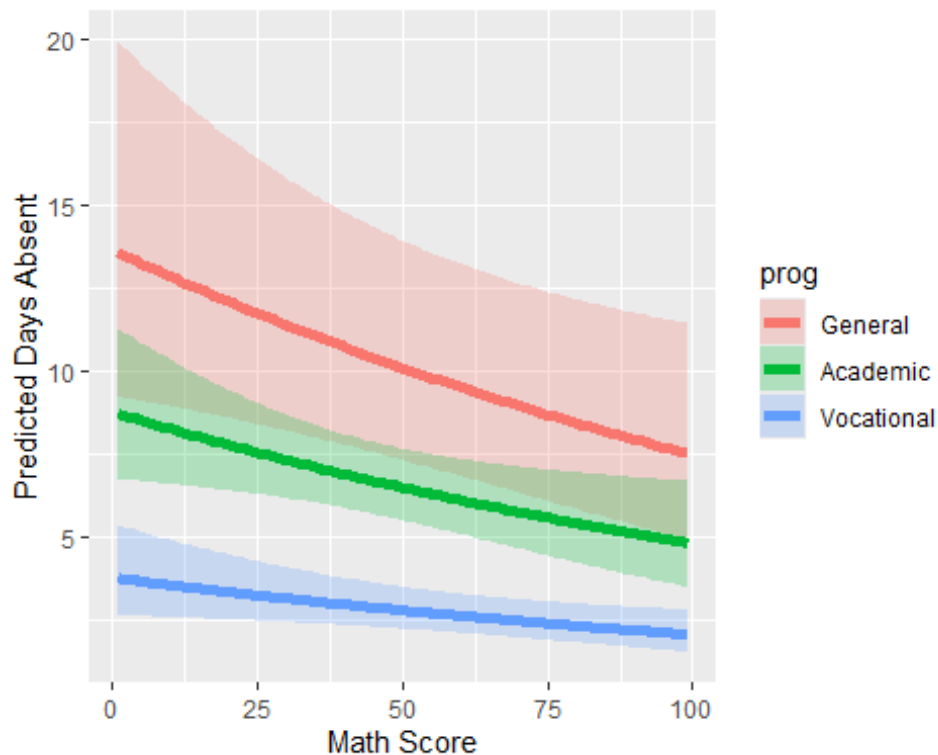
R code and results: Graphical representation

```
####graph
newdata2 <- data.frame(
  math = rep(seq(from = min(dat$math), to = max(dat$math), length.out = 10
0), 3),
  prog = factor(rep(1:3, each = 100), levels = 1:3, labels =
levels(dat$prog)))

newdata2 <- cbind(newdata2, predict(m1, newdata2, type = "link", se.fit=TR
UE))
newdata2 <- within(newdata2, {
  DaysAbsent <- exp(fit)
  LL <- exp(fit - 1.96 * se.fit)
  UL <- exp(fit + 1.96 * se.fit)
})

ggplot(newdata2, aes(math, DaysAbsent)) +
  geom_ribbon(aes(ymin = LL, ymax = UL, fill = prog), alpha = .25) +
```

```
geom_line(aes(colour = prog), size = 2) +  
labs(x = "Math Score", y = "Predicted Days Absent")
```



Comment: The above graph shows the expected count across the range of math scores, for each type of program along with 95 percent confidence intervals. Here the lines are not straight because this is a log linear model, and what is plotted are the expected values, not the log of the expected values.

[Zero Inflated Negative Binomial Distribution](#)

Question 4: The state wildlife biologists want to model how many fish are being caught by fishermen at a state park. Visitors are asked how long they stayed, how many people were in

the group, were there children in the group and how many fish were caught. Some visitors do not fish, but there is no data on whether a person fished or not. Some visitors who did fish did not catch any fish so there are excess zeros in the data because of the people that did not fish.

R code and results:

```
###zero inflated
install.packages("pscl")

install.packages("boot")

library(ggplot2)
library(pscl)
library(boot)
#
zinb <- read.csv("https://stats.idre.ucla.edu/stat/data/fish.csv")
zinb <- within(zinb, {
  nofish <- factor(nofish)
  livebait <- factor(livebait)
  camper <- factor(camper)
})

summary(zinb)
```

##	nofish	livebait	camper	persons	child	xb
zg						
##	0:176	0: 34	0:103	Min. :1.000	Min. :0.000	Min. : -3.27
5050	Min. : -5.6259					
##	1: 74	1:216	1:147	1st Qu.:2.000	1st Qu.:0.000	1st Qu.: 0.00
8267	1st Qu.: -1.2527					
##				Median :2.000	Median :0.000	Median : 0.95
4550	Median : 0.6051					
##				Mean :2.528	Mean :0.684	Mean : 0.97
3796	Mean : 0.2523					
##				3rd Qu.:4.000	3rd Qu.:1.000	3rd Qu.: 1.96
3855	3rd Qu.: 1.9932					
##				Max. :4.000	Max. :3.000	Max. : 5.35
2674	Max. : 4.2632					
##	count					
##	Min. : 0.000					
##	1st Qu.: 0.000					
##	Median : 0.000					
##	Mean : 3.296					
##	3rd Qu.: 2.000					
##	Max. :149.000					

Comment: The summary result shows some descriptive study of the data. There is minimum 1 person and maximum 4 person in each group. There is minimum 0 children and maximum 3 three children there in each group.

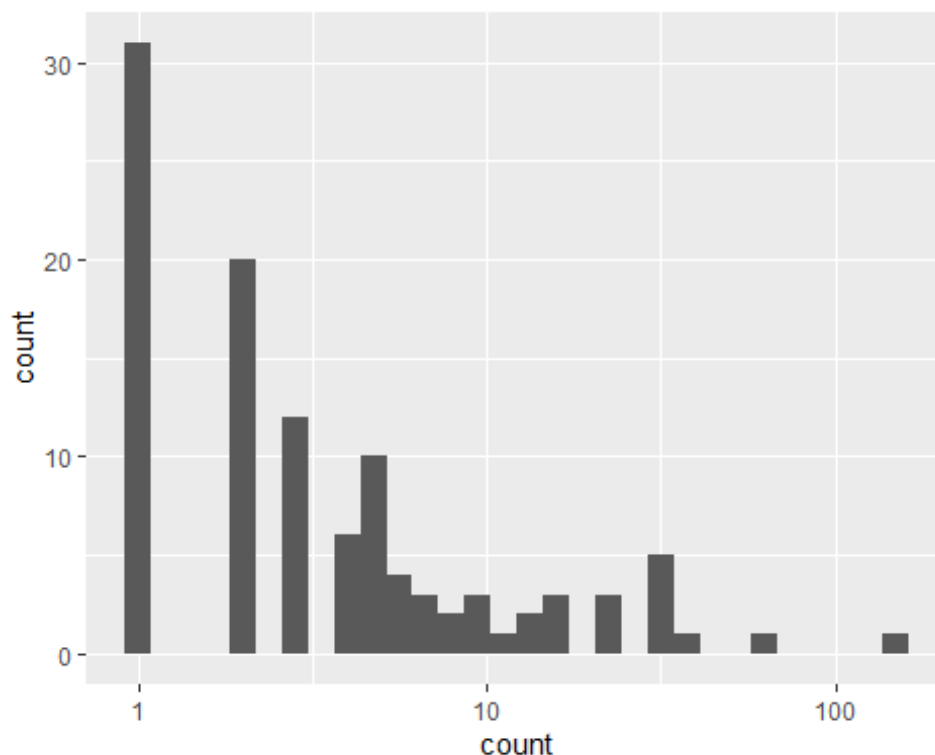
R code and results:

```
## histogram with x axis in log10 scale
ggplot(zinb, aes(count)) + geom_histogram() + scale_x_log10()

## Warning in scale_x_log10(): log-10 transformation introduced infinite v
alues.

## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.

## Warning: Removed 142 rows containing non-finite outside the scale range
(`stat_bin()`).
```



Comment: The above graph is a histogram with x axis in log10 scale.

R code and results:

```
summary(m1 <- zeroinfl(count ~ child + camper | persons, data = zinb))

##
## Call:
## zeroinfl(formula = count ~ child + camper | persons, data = zinb)
##
## Pearson residuals:
##      Min      1Q  Median      3Q      Max
## -1.2369 -0.7540 -0.6080 -0.1921 24.0847
##
## Count model coefficients (poisson with log link):
```

```
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.59789    0.08554  18.680  <2e-16 ***
## child       -1.04284    0.09999 -10.430  <2e-16 ***
## camper1      0.83402    0.09363   8.908  <2e-16 ***
##
## Zero-inflation model coefficients (binomial with logit link):
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.2974    0.3739   3.470 0.000520 ***
## persons     -0.5643    0.1630  -3.463 0.000534 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Number of iterations in BFGS optimization: 12
## Log-likelihood: -1032 on 5 Df
```

Comment: The above outcome illustrates that when all predictors are zero, the expected count is approximately 1.60. For each unit increase in child, the expected count decreases by approximately 1.04. Being a camper increases the expected count by approximately 0.83. The log-odds of excess zeros is approximately 1.30 when all other predictors are zero. For each unit increase in persons, the log-odds of excess zeros decreases by approximately 0.56. Hence we can say, child and camper significantly influence the count of the outcome. Persons significantly affects the likelihood of excess zeros.

R code and results:

```
#
mnull <- update(m1, . ~ 1)

pchisq(2 * (logLik(m1) - logLik(mnull)), df = 3, lower.tail = FALSE)

## 'log Lik.' 4.041242e-41 (df=5)
```

Comment: The result indicates an extremely low p-value (**4.041242e-41**), suggesting strong evidence against the null hypothesis (the null model), indicating that the predictors in the original model are collectively significant in explaining the outcome.

R code and results:

```
dput(coef(m1, "count"))

## c`(Intercept)` = 1.59788828690411, child = -1.04283909332231,
## camper1 = 0.834023618148891)
```

```
dput(coef(m1, "zero"))

## c(`(Intercept)` = 1.29744027908309, persons = -0.564347365357873
## )

#
f <- function(data, i) {
  require(psc1)
  m <- zeroinfl(count ~ child + camper | persons, data = data[i, ],
    start = list(count = c(1.598, -1.0428, 0.834), zero = c(1.
297, -0.564)))
  as.vector(t(do.call(rbind, coef(summary(m)))[, 1:2])))
}

set.seed(10)
res <- boot(zinb, f, R = 1200, parallel = "snow", ncpus = 4)

#print results
res

##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = zinb, statistic = f, R = 1200, parallel = "snow",
##      ncpus = 4)
##
##
## Bootstrap Statistics :
##      original      bias    std. error
## t1*    1.59788854 -0.0451465197  0.29947818
## t2*    0.08553816  0.0036239299  0.01659791
## t3*   -1.04283849  0.0112733726  0.41120836
## t4*    0.09998829  0.0042049207  0.01580662
## t5*    0.83402218  0.0003032759  0.40949791
## t6*    0.09362679  0.0041210986  0.01524233
## t7*    1.29743916  0.0211989505  0.47396484
## t8*    0.37385225  0.0069068079  0.03657407
## t9*   -0.56434716 -0.0265559374  0.26840571
## t10*   0.16296381  0.0039579583  0.03019734
```

Comment: The above results are alternating parameter estimates and standard errors. That is, the first row has the first parameter estimate from our model. The second has the standard error for the first parameter. The third column contains the bootstrapped standard errors, which are considerably larger than those estimated by zeroinfl.

R code and results:

```
## basic parameter estimates with percentile and bias adjusted CIs
parms <- t(sapply(c(1, 3, 5, 7, 9), function(i) {
```

```

out <- boot.ci(res, index = c(i, i + 1), type = c("perc", "bca"))
with(out, c(Est = t0, pLL = percent[4], pUL = percent[5],
            bcaLL = bca[4], bcaLL = bca[5]))
)))
parms

##              Est              pLL              pUL              bcaLL              bcaLL
## [1,]  1.5978885  0.929548809  2.0891957  1.052591464  2.21402500
## [2,] -1.0428385 -1.778733547 -0.1686651 -1.727480382 -0.09541710
## [3,]  0.8340222  0.008795793  1.5978058  0.001069699  1.58913696
## [4,]  1.2974392  0.458982712  2.2596422  0.441060605  2.21684371
## [5,] -0.5643472 -1.100952918 -0.1167136 -1.061884331 -0.03569355

## add row names
row.names(parms) <- names(coef(m1))
## print results
parms

##              Est              pLL              pUL              bcaLL
bcaLL
## count_(Intercept)  1.5978885  0.929548809  2.0891957  1.052591464  2.21
402500
## count_child        -1.0428385 -1.778733547 -0.1686651 -1.727480382 -0.09
541710
## count_camper1       0.8340222  0.008795793  1.5978058  0.001069699  1.58
913696
## zero_(Intercept)   1.2974392  0.458982712  2.2596422  0.441060605  2.21
684371
## zero_persons       -0.5643472 -1.100952918 -0.1167136 -1.061884331 -0.03
569355

## compare with normal based approximation
confint(m1)

##              2.5 %              97.5 %
## count_(Intercept)  1.4302366  1.7655400
## count_child        -1.2388125 -0.8468657
## count_camper1       0.6505185  1.0175288
## zero_(Intercept)   0.5647033  2.0301772
## zero_persons       -0.8837505 -0.2449442

```

Comment: From the above results we can say, the bootstrapped confidence intervals are considerably wider than the normal based approximation.

R code and results:

```

## exponentiated parameter estimates with percentile and bias adjusted CIs
expparms <- t(sapply(c(1, 3, 5, 7, 9), function(i) {

```

```

out <- boot.ci(res, index = c(i, i + 1), type = c("perc", "bca"), h = ex
p)
with(out, c(Est = t0, pLL = percent[4], pUL = percent[5],
            bcaLL = bca[4], bcaUL = bca[5]))
)))

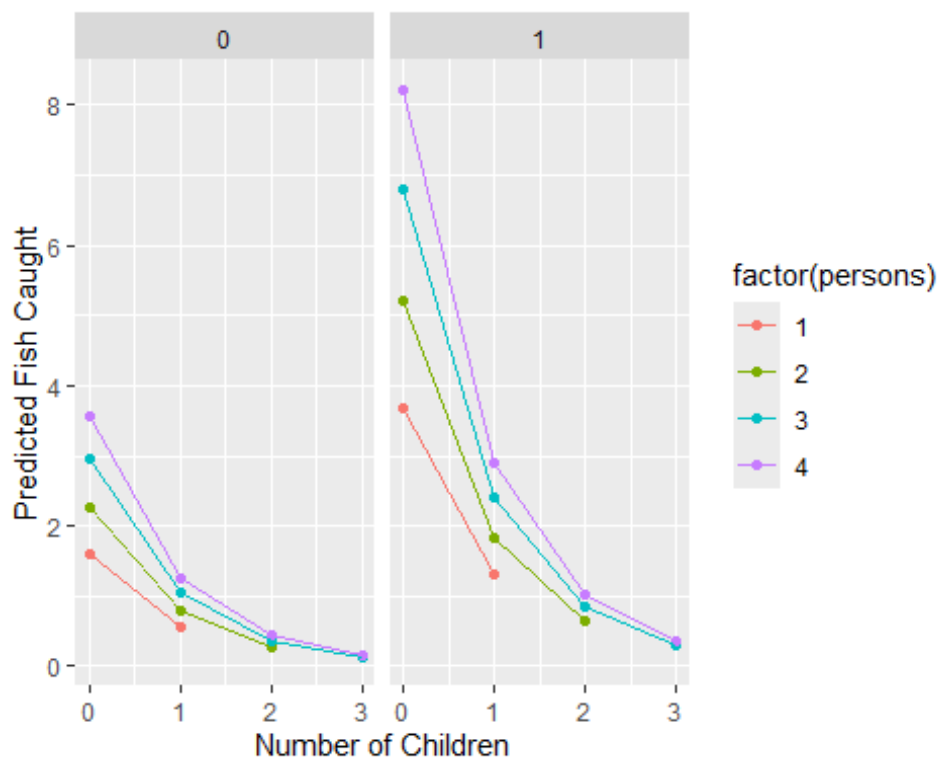
## add row names
row.names(expparms) <- names(coef(m1))
## print results
expparms

##
##              Est          pLL          pUL          bcaLL          bcaUL
## count_(Intercept) 4.9425853 2.5333659 8.0784160 2.8650662 9.1524811
## count_child       0.3524528 0.1688519 0.8447918 0.1777317 0.9089937
## count_camper1     2.3025614 1.0088346 4.9421772 1.0010703 4.8995186
## zero_(Intercept)  3.6599122 1.5824634 9.5796696 1.5543549 9.1783157
## zero_persons      0.5687313 0.3325540 0.8898401 0.3458036 0.9649360

##
newdata1 <- expand.grid(0:3, factor(0:1), 1:4)
colnames(newdata1) <- c("child", "camper", "persons")
newdata1 <- subset(newdata1, subset=(child<=persons))
newdata1$phat <- predict(m1, newdata1)

ggplot(newdata1, aes(x = child, y = phat, colour = factor(persons))) +
  geom_point() +
  geom_line() +
  facet_wrap(~camper) +
  labs(x = "Number of Children", y = "Predicted Fish Caught")

```



Comment: The x axis represents the number of children in a family and the y axis represents the predicted number of fish caught. The lines on the plot represent the trend or relationship between the number of children and the predicted number of fish caught. They provide a visual representation of how the predicted number of fish caught changes as the number of children changes. The plot is divided into separate panels based on the camper variable. Each panel shows the relationship between child and phat for a specific level of camper. This allows for easier comparison between different camper categories.